

Soohyeon Choi

Research Scientist | AI and LLM Safety and Security

Singapore shchoi@smu.edu.sg soohyeonc.github.io Google Scholar
github.com/soohyeonc linkedin.com/in/soohyeon-choi ORCID 0009-0002-1252-2263

Profile

AI and LLM security researcher with a Ph.D. in Computer Science and more than five years of research experience in adversarial evaluation, generated-code provenance, jailbreak analysis, code watermarking, and robust machine learning. Designs reproducible benchmarks, fine-tunes code models, diagnoses model failure under distribution shift, and publishes across AI, software, and security venues. Research has received 295 Google Scholar citations as of August 2026.

Research Experience

Singapore Management University

Research Scientist I

Singapore

May 2025–Present

- Conduct research on AI and LLM security, including compositional jailbreak attacks, post-hoc code watermarking, generated-code provenance, and attribution of code produced by AI coding agents.
- Design large-scale evaluation pipelines using fine-tuned code encoders, zero-shot detectors, robustness tests, matched controls, and statistically audited datasets.
- Collaborate with academic and industry researchers across Singapore and the United States, and communicate findings through manuscripts, artifacts, and research presentations.

University of Central Florida

Graduate Research Assistant, SEAL Lab

Orlando, Florida, USA

Aug. 2021–May 2025

- Developed machine-learning methods to identify ChatGPT-generated and ChatGPT-transformed source code and evaluated their robustness to code transformation and distribution shift.
- Built a sequence-to-sequence code-transformation approach for authorship evasion and studied adversarial robustness in language and sensor-based classification systems.
- Published first-authored work in IEEE TDSC, IEEE ICDCS, ICICS, and CSoNet while completing a Ph.D. focused on machine learning, LLMs, and security.

A*STAR Institute for Infocomm Research

Research Officer

Singapore

Dec. 2023–Dec. 2024

- Investigated zero-shot and few-shot LLM methods for source-code authorship attribution, including cross-language generalization, adversarial transformation, and security implications.

South Dakota State University

Graduate Research Assistant, CCT Lab

Brookings, South Dakota, USA

Aug. 2019–Dec. 2020

- Developed a low-computation authentication protocol for wireless body-area networks using elliptic-curve cryptography and two-dimensional hash chains.

Education

Ph.D. in Computer Science, University of Central Florida

2021–2025

Advisor: Prof. David Mohaisen. Focus: machine learning, LLMs, and security.

Orlando, USA

M.S. in Computer Science, South Dakota State University

2018–2021

Advisor: Prof. Sung Shin. Focus: authentication and sensor-network security.

Brookings, USA

B.E. in Computer Engineering, Keimyung University

2011–2017

Daegu, South Korea

Selected AI Safety and Security Research

Compositional Jailbreaking of Aligned LLMs

Corresponding author; under review, 2026

- Evaluated all ordered pairs of 12 jailbreak mutators against three LLMs to identify synergistic, destructive, and structurally incompatible attack compositions; developed completeness and validity measures for attack persistence and effectiveness.

CommitTrace: Fine-Grained Attribution of AI Coding Agents

Lead research project; under review, 2026

- Diagnosed why prior detectors fail on repository-local changes, with zero-shot AUROC ranging from 0.398 to 0.528, then developed a code-only representation combining a fine-tuned encoder, character n-grams, and stylometry.
- Achieved binary macro F1 of 0.966 on file patches and 0.953 on hunks across five coding agents; evaluated fragmentation, context, robustness, cross-language transfer, and repository-disjoint generalization.

Multi-Channel Spread-Spectrum Code Watermarking

First author; under review, 2026

- Developed a post-hoc, training-free method that embeds a 24-bit provider label through naming and structural channels with explicit erasure-oriented recovery.
- Evaluated 1,750 Python files with 100% clean target-label recovery and 94.1% recovery under 10% random per-site corruption in the tested setting; embedding and detection run on CPU without access to the generating model.

Selected Peer-Reviewed Publications

- **S. Choi** and D. Mohaisen, “Attributing ChatGPT-Generated Source Codes,” *IEEE Transactions on Dependable and Secure Computing*, 2025.
- **S. Choi** et al., “Attributing ChatGPT-Transformed Synthetic Code,” *IEEE International Conference on Distributed Computing Systems*, 2025.
- M. F. Kharma, **S. Choi**, M. Alkhanafseh, and D. Mohaisen, “Security and Quality in LLM-Generated Code: A Multi-Language, Multi-Model Analysis,” *IEEE Transactions on Dependable and Secure Computing*, 2026.
- A. Alkinoon, T. C. Dang, A. Alghuried, A. Alghamdi, **S. Choi**, et al., “A Comprehensive Analysis of Evolving Permission Usage in Android Apps: Trends, Threats, and Ecosystem Insights,” *Journal of Cybersecurity and Privacy*, 2025.
- **S. Choi**, R. Jang, D. Nyang, and D. Mohaisen, “Untargeted Code Authorship Evasion with Seq2Seq Transformation,” *CSoNet*, 2023.
- **S. Choi**, M. Mohaisen, D. Nyang, and D. Mohaisen, “Revisiting the Deep Learning-Based Eavesdropping Attacks via Facial Dynamics from VR Motion Sensors,” *ICICS*, 2023.
- M. Omar, **S. Choi**, D. Nyang, and D. Mohaisen, “Quantifying the Performance of Adversarial Training on Language Models with Distribution Shifts,” *CySSS*, 2022.
- M. Omar, **S. Choi**, D. Nyang, and D. Mohaisen, “Robust Natural Language Processing: Recent Advances, Challenges, and Future Directions,” *IEEE Access*, 2022.

Selected Manuscripts Under Review and Preprint

- **S. Choi** et al., “CommitTrace for Code-Only Attribution of AI Coding Agents from File Patches and Hunks,” under review, 2026.
- **S. Choi**, D. Gao, and Y. Duan, “Multi-Channel Spread-Spectrum Code Watermarking,” under review; arXiv:2607.06009, 2026.
- R. J. Bugnot, **S. Choi**, H. W. Lim, and Y. Duan, “Compositional Jailbreaking: An Empirical Analysis of Mutator Chain Interactions in Aligned LLMs,” under review; arXiv:2605.15598, 2026. *Corresponding author*.
- **S. Choi** et al., “I Can Find You in Seconds! Leveraging Large Language Models for Code Authorship Attribution,” preprint; arXiv:2501.08165, 2025.

Technical Skills and Professional Service

Programming and tools: Python, C/C++, Java, SQL, Linux, Git, L^AT_EX.

Machine learning: PyTorch, TensorFlow, scikit-learn, Transformers, model fine-tuning, LLM evaluation, adversarial ML, benchmark design, statistical evaluation.

Reviewing: IEEE TIFS (2025–2026), IEEE TDSC (2023–2026), ACM AsiaCCS (2025), IEEE TMC (2022), and *Computer Networks* (2022).